



Statistical Techniques for Computer Applications

Unit No. 2 : Measures of Central Tendency and Measures of Dispersion

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Syllabus

Unit II: Measures of Central Tendency and Measures of Dispersion

Measures of Central Tendency

-Mean, Median and Mode for Continuous data, Comparison of Mean, Median and Mode as Measures of Central Tendency

Measures of Dispersion

-Range, Quartile Deviation, Standard Deviation, Variance and Coefficient of Variation for Continuous data.



Measures of Central Tendency

-Mean, Median and Mode for Continuous data, Comparison of Mean, Median and Mode as Measures of Central Tendency



Measures of central tendency

Measures of central tendency are statistical tools used to represent a large set of data by a single central value.

In computer applications, they help in data analysis, system evaluation, performance monitoring, and decision making.

The main measures are:

- Mean
- Median
- Mode

Mean

Definition

Mean is the average value obtained by dividing the sum of all observations by the number of observations.

Uses of Mean

- To analyze average system performance
- Used in CPU usage, memory utilization, response time
- Helpful in capacity planning

Example:

Average server response time over a week.

Advantages

- Easy to calculate
- Uses all observations
- Suitable for uniform data

Where Mean is Useful:

- Measuring average bandwidth usage
- Average query execution time
- Average app download per day
- Average coding test score
- Average bug resolution time

Median

Definition

Median is the middle value of a dataset when observations are arranged in ascending or descending order.

Uses of Median

- Used when performance data has extreme values (server spikes)
- Helpful in analyzing response times
- Used in database query performance analysis
- Useful in network latency measurement
- Used in academic performance analysis of coding tests

Examples

- Median server response time of a web application over a week
-
- Median execution time of an algorithm tested multiple times
 - Median network latency in milliseconds during peak hours
 - Median marks in programming practical exam
 - Median database query processing time
 - Median app load time recorded from different devices
 - Median CPU usage percentage during system testing

Advantages

- Not affected by extreme spikes (like sudden server overload)
- Better measure for skewed system performance data
- Gives realistic central performance value
- Useful in performance benchmarking
- Suitable for ordered data analysis

Mode

Definition

Mode is the value that occurs most frequently in a dataset.

Uses of Mode

- Identifies most commonly used technology
- Useful in user behavior analysis
- Used in software usage statistics
- Helpful in market research for tech products
- Used in system error analysis

Examples

- Most used programming language in a company
- Most common error code in server logs
- Most frequently accessed webpage in a website
- Most common mobile OS version used by users
- Most selected database management system in projects
- Most frequently purchased software subscription plan
- Most common browser used by website visitors

Advantages

- Best for categorical data (like programming languages)
- Easy to identify in logs and datasets
- Helps in product and feature planning
- Useful in trend detection
- Supports business and technical decision-making

Role of Central Tendency (Mean, Median, Mode) in Computer Applications

Central tendency measures — Mean, Median, and Mode — are statistical tools used to summarize large datasets into a single representative value. In computer applications, massive volumes of data are generated through systems, networks, databases, and user interactions. These measures help in analysis, optimization, and informed decision-making.

Role of Central Tendency (Mean, Median, Mode) in Computer Applications

1. System Performance Analysis

Software systems continuously generate performance metrics such as CPU usage, memory utilization, and response time.

Mean → Measures average server response time or CPU usage.

Median → Shows typical response time by ignoring extreme spikes.

Mode → Identifies the most frequent error code or failure type.

Examples:

- Mean API response time of a web application
- Median database transaction time during peak hours
- Mode of most common HTTP status code (e.g., 404)
- Mean disk I/O speed
- Median virtual machine boot time

Helps in system monitoring, load balancing, and performance tuning.

Role of Central Tendency (Mean, Median, Mode) in Computer Applications

2. Database Management Systems (DBMS)

Large databases handle millions of transactions daily.

Mean query execution time evaluates overall database efficiency.

Median transaction time reflects typical system behavior.

Mode of transaction type identifies most frequent operations.

Examples:

- Mean data retrieval time from cloud storage
- Median record insertion time
- Mode of most accessed table
- Mean number of transactions per hour
- Median backup completion time

Supports indexing, caching strategies, and query optimization.

Role of Central Tendency (Mean, Median, Mode) in Computer Applications

3. Network Monitoring and Security

Network administrators analyze latency, bandwidth, and error rates.

Mean bandwidth usage indicates overall network load.

Median latency avoids distortion caused by traffic spikes.

Mode of network error identifies frequent connectivity issues.

Examples:

- Mean packet transmission delay
- Median ping time to remote servers
- Mode of most common firewall alert
- Mean download speed
- Median upload speed

Assists in troubleshooting, capacity planning, and network optimization.

Role of Central Tendency (Mean, Median, Mode) in Computer Applications

4. Machine Learning & Data Analytics

Before model development, data must be summarized and cleaned.

Mean is used for normalization and feature scaling.

Median is used to replace missing numerical values.

Mode is used to fill missing categorical values.

Examples:

- Mean pixel intensity in image processing
- Median income value in a dataset
- Mode of most common class label
- Mean model accuracy score
- Median prediction error

Essential for data preprocessing and improving model accuracy.



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5. User Behavior & Web Analytics

Websites and applications track user interactions.

Mean session duration shows average engagement.

Median page load time reflects realistic user experience.

Mode of most visited page identifies popular content.

Examples:

- Mean number of clicks per user
- Median checkout time
- Mode of most used feature in an app
- Mean daily active users
- Mode of most common login time

Helps in UI/UX improvement and business strategy.

6. Academic & Software Evaluation

Central tendency is also useful in academic and project evaluation.

Examples:

- Mean marks in programming exams
- Median project evaluation scores
- Mode of most selected programming language
- Mean coding test completion time
- Median internship performance rating

Helps in performance assessment and curriculum planning.

Conclusion

In computer applications, Mean, Median, and Mode play a crucial role in:

- Summarizing large datasets
- Monitoring and improving system performance
- Supporting data-driven decision making
- Optimizing software and network resources
- Enhancing business intelligence and analytics

Thus, central tendency measures are fundamental tools in modern computing and data-driven environments.

Example 1

Response time (in ms) of a web server is grouped as follows:

Response Time (ms)	Frequency
100 – 200	6
200 – 300	14
300 – 400	20
400 – 500	8
500 – 600	2

Question:

Q. 1 From the above data Calculate the mean response time, Median and Mode

Q. 2 Decide whether the server meets the performance target of 350 ms

Solution :

Step 1: Find class midpoints (x)

Class	Midpoint (x)	f	fx
100–200	150	6	900
200–300	250	14	3500
300–400	350	20	7000
400–500	450	8	3600
500–600	550	2	1100

Solution

$$\sum f = 50, \quad \sum fx = 16100$$

$$\text{Mean} = \frac{\sum fx}{\sum f} = \frac{16100}{50} = 322 \text{ ms}$$

Mean response time = 322 ms

Q. 2 Decide whether the server meets the performance target of 350 ms

Decision Making

Answer :

Mean response time = 322 ms, which is below the target of 350 ms.

Decision:

Server performance is acceptable, and no immediate optimization is required.

Example 2

The monthly data usage (in GB) of 100 company users is given below:

Data Usage (GB)	Frequency (f)
100 – 200	5
200 – 300	12
300 – 400	20
400 – 500	28
500 – 600	18
600 – 700	10
700 – 800	7

Questions:

Q.1 Calculate Mean, Median and Mode for the above data.

Q. 2 Should the company design a 450 GB data plan as a standard corporate plan? Justify using mean and median.

Q.3 Should server capacity be increased beyond 500 GB per user considering central tendency values?

Q.4 Which measure best represents the “typical” user’s data consumption and why?

Q. 5 If management wants to introduce a premium data plan, which class interval should be targeted based on modal class?

Solution : Calculation of Mean

Step 1: Calculate Midpoints (m)

Class Interval	f	Midpoint (m)	fm
100–200	5	150	750
200–300	12	250	3000
300–400	20	350	7000
400–500	28	450	12600
500–600	18	550	9900
600–700	10	650	6500
700–800	7	750	5250

$$\text{Mean} = \frac{\sum fm}{N}$$

$$\sum fm = 45000$$

$$\text{Mean} = \frac{45000}{100} = 450$$

Mean = 450 GB

Calculation of Median

Calculation of Cumulative Frequency

Class	f	C.F.
100–200	5	5
200–300	12	17
300–400	20	37
400–500	28	65
500–600	18	83
600–700	10	93
700–800	7	100

$$N/2 = 100/2 = 50$$

50 lies in the class **400–500** → Median Class

Formula

$$\text{Median} = l_1 + \left(\frac{\frac{N}{2} - CF}{f} \right) \times (l_2 - l_1)$$

Where:

- l_1 = Lower limit of median class
- l_2 = Upper limit of median class
- N = Total frequency
- CF = Cumulative frequency before median class
- f = Frequency of median class



Substitute

$$\begin{aligned}\text{Median} &= 400 + \left(\frac{50 - 37}{28} \right) \times (500 - 400) \\ &= 400 + \left(\frac{13}{28} \right) \times 100 \\ &= 400 + 46.43\end{aligned}$$

$$\text{Median} \approx 446.43 \text{ GB}$$



Calculation of Mode

$$\text{Mode} = L + \frac{d_1}{d_1 + d_2} \times (L_2 - L_1)$$

Modal class = 400–500 (Highest frequency = 28)

$$d_1 = f_1 - f_0 = 28 - 20 = 8$$

$$d_2 = f_1 - f_2 = 28 - 18 = 10$$

$$\begin{aligned}\text{Mode} &= 400 + \frac{8}{8 + 10} \times 100 \\ &= 400 + 44.44\end{aligned}$$

Mode $\approx$ 444.44 GB



Answer :

Q2. Should the company design a 450 GB data plan as a standard corporate plan?

Answer :

Yes, designing a 450 GB plan is reasonable.

Mean = 450 GB → Average usage.

Median $\approx$ 446 GB → Typical user consumes around 446 GB.

Since both mean and median are close to 450 GB, it represents the central consumption level.

Conclusion:

A 450 GB standard plan would satisfy the majority of users.

Answer

Q.3 Should server capacity be increased beyond 500 GB per user considering central tendency values?

Answer:

No — the server capacity should NOT be increased beyond 500 GB per user.

Reason:

Average user usage $\approx$ 450 GB

Majority of users fall in 400–500 GB range

500 GB already provides a safety buffer of $\sim$ 50 GB

Increasing capacity would lead to unnecessary infrastructure cost without real demand.

Answer :

Q4. Which measure best represents the “typical” user’s data consumption and why?

Answer :

The Median best represents the typical user.

Reason:

It divides data into two equal halves.

It is not influenced by extreme users.

It reflects central user behavior.

Conclusion: Median $\approx$ 446 GB is the best representation.

Answer :

Q5. If management wants to introduce a premium data plan, which class interval should be targeted?

Answer :

Premium users are in higher consumption ranges:

500–600 GB

600–700 GB

700–800 GB

Since modal class is 400–500 (standard users), premium plan should target:

500–600 GB or above

Conclusion:

Premium plan should start from 500 GB+ category.

Example 3

ABC Tech Solutions (a cloud services company) analyzed CPU utilization (%) of its production servers over 200 peak-hour observations.

CPU Usage (%)	Frequency (f)
0–20	10
20–40	30
40–60	60
60–80	70
80–100	30

Questions

Q.1 Calculate Mean, Median and Mode for the above data.

Q. 2 Is current infrastructure underutilized?

Q. 3 Should the company upgrade server capacity?

Q4. Should the company immediately upgrade server infrastructure?

Q5. Should the company reduce servers to save operational cost?

Q. 6 Why is median (60%) important in this project report?

CALCULATION OF MEAN

Step 1: Find Midpoints

Class	f	Midpoint (m)	f × m
0–20	10	10	100
20–40	30	30	900
40–60	60	50	3000
60–80	70	70	4900
80–100	30	90	2700

$$\bar{X} = \frac{\sum fm}{N}$$

$$\bar{X} = \frac{11600}{200}$$

$$\bar{X} = 58$$

Calculation of Median

Step 1: Cumulative Frequency

Class	f	Cumulative f
0–20	10	10
20–40	30	40
40–60	60	100
60–80	70	170
80–100	30	200

$$N/2 = 100$$

Median class = **40–60** (since cumulative frequency reaches 100)

Formula

$$\text{Median} = l_1 + \left(\frac{\frac{N}{2} - CF}{f} \right) \times (l_2 - l_1)$$

Where:

- l_1 = Lower limit of median class
- l_2 = Upper limit of median class
- N = Total frequency
- CF = Cumulative frequency before median class
- f = Frequency of median class



Substitute Values in the Formula

So,

- $L = 40$
- $cf = 40$ (cumulative before median class)
- $f = 60$
- $h = 20$

$$\begin{aligned} \text{Median} &= 40 + \left(\frac{100 - 40}{60} \right) \times 20 \\ &= 40 + \left(\frac{60}{60} \right) \times 20 \\ &= 40 + 20 \\ &= 60 \end{aligned}$$

Median CPU Usage = 60%

Calculation of Mode

Modal class = **60–80**

Because it has highest frequency = 70

$$\text{Mode} = L + \frac{d_1}{d_1 + d_2} \times (L_2 - L_1)$$

- $L = 60$ (lower limit of modal class)
- $L_1 = 60$
- $L_2 = 80$
- $d_1 = f_m - f_1 = 70 - 60 = 10$
- $d_2 = f_m - f_2 = 70 - 30 = 40$

$$\begin{aligned} \text{Mode} &= 60 + \frac{10}{10 + 40} \times (80 - 60) \\ &= 60 + \frac{10}{50} \times 20 \\ &= 60 + 0.2 \times 20 \\ &= 60 + 4 \\ &= 64 \end{aligned}$$

Answer : Q. 1

Mean = 58%

Median = 60%

Mode = 64%



Answer :

Q. 2 Is current infrastructure underutilized?

Answer : No. the current infrastructure is not underutilized. Most servers are operating above 50%.

The **mean CPU usage is 58%**, which indicates that on average servers are operating at moderate load.

The **median is 60%**, meaning 50% of the peak-hour observations are at or above 60% utilization.

The **mode is 64%**, showing that the most frequently occurring CPU usage lies in the higher utilization range (60–80%).

Based on these statistical measures, the servers are consistently operating at moderate to high utilization levels. Therefore, the infrastructure cannot be considered underutilized.

Answer :

Q. 3 Should the company upgrade server capacity?

Answer :

Since median = 60% and modal load $\approx$ 64%, system is moderately stressed.

Future scaling recommended.

Q4. Should the company immediately upgrade server infrastructure?

Answer:

Immediate full upgrade is not mandatory, but preventive scaling should begin.

Although mean CPU usage is 58%, 50% of observations lie above 60%.

Mode = 64% indicates frequent high utilization.

This suggests growing demand and reduced performance buffer.

A phased horizontal scaling strategy is recommended.

Answer :

Q5. Should the company reduce servers to save operational cost?

Answer:

No.

Only 5% observations lie in 0–20% range, meaning servers are rarely idle.

Most data lies between 40–80%, showing efficient but heavy usage.

Reducing servers may push system into overload zone (80–100%).

Cost-cutting here increases risk.

Q. 6 Why is median (60%) important in this project report?

Answer :

Median shows central operating level.

Since median = 60%, half of peak hours operate at or above 60%.

This confirms consistent moderate-to-high usage.

Practice Example 4

A software company analyzed the **session duration (in minutes)** of 120 users visiting its web application during peak hours. The data was grouped as follows:

Session Duration (minutes)	Number of Users
0–10	8
10–20	22
20–30	40
30–40	28
40–50	15
50–60	7

Comparison of Mean, Median and Mode as Measures of Central Tendency



Comparison of Mean, Median and Mode as Measures of Central Tendency

$$\text{Mode} = \text{Mean} - 3(\text{Mean} - \text{Median})$$

$$\text{Mode} = 3 \text{ Median} - 2\text{mean}$$

$$\text{Median} = \text{Mode} + \frac{2}{3} (\text{Mean} - \text{Mode})$$

Why can we calculate Mode using Mean and Median without knowing the full dataset?

Because in a moderately skewed distribution, there exists an empirical relationship:

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

This formula estimates mode based on central tendency behavior.

However, it is only valid when the distribution is approximately symmetric or moderately skewed.

When

$$\text{Mean} = \text{Median} = \text{Mode}$$

This indicates a perfectly symmetric distribution.

Example 1 :

A cloud computing company analyzed the server response time (in milliseconds) during peak hours for its web application.

After statistical analysis:

- The mean response time was found to be 240 ms.
- The median response time was found to be 220 ms.

Assuming the distribution is moderately skewed, calculate the mode of response time.

Solution

Using the empirical formula:

$$Mode = 3Median - 2Mean$$

Substitute values:

$$\begin{aligned} Mode &= 3(220) - 2(240) \\ &= 660 - 480 \\ &= 180 \end{aligned}$$



Interpretation :

- Mean (240 ms) is highest → influenced by some very high response times (slow servers).
- Median (220 ms) shows central performance.
- Mode (180 ms) indicates the most frequently occurring response time is lower.

Since:

Mode < Median < Mean

This indicates **positive skewness**, meaning some high response-time spikes are pulling the average upward.

Example 2

A cloud storage company studied daily server load (%).

The most frequent load (Mode) = 55%

Median load = 60%

Find the Mean server load.

Solution :

$$\text{Mean} = \frac{3\text{Median} - \text{Mode}}{2}$$

$$\text{Mean} = \frac{3(60) - 55}{2}$$

$$= \frac{180 - 55}{2}$$

$$= \frac{125}{2}$$

$$= 62.5$$

$$\text{Mean} = 62.5\%$$



Example 2

A cloud hosting company analyzed page load time (in milliseconds) during peak hours.

After statistical analysis:

Mean page load time = 250 ms

The difference between Mean and Median is 10 ms

Find:

1. Median
2. Mode

Solution

$$\text{Mean} - \text{Median} = 10$$

$$\text{Median} = 250 - 10 = 240$$

Using empirical formula:

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

$$\begin{aligned}\text{Mode} &= 3(240) - 2(250) \\ &= 720 - 500 = 220\end{aligned}$$

Hence

$$\text{Mean} = 250$$

$$\text{Median} = 240$$

$$\text{Mode} = 220$$

Example 3

An IT firm monitored network latency (in milliseconds) for its data center.

After analysis:

Mean latency = 95 ms

Mean exceeds Median by 5 ms

Find:

1. Median latency
2. Mode latency

Solution

$$\text{Mean} - \text{Median} = 5$$

$$95 - \text{Median} = 5$$

$$\text{Median} = 90 \text{ ms}$$

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

$$\begin{aligned}\text{Mode} &= 3(90) - 2(95) \\ &= 270 - 190 = 80\end{aligned}$$

Measures of Dispersion

- Range,
- Quartile Deviation,
- Standard Deviation,
- Variance and Coefficient of Variation for Continuous data.



Measures of Dispersion

Measures of dispersion are statistical measures that indicate the extent to which individual observations in a dataset deviate or spread out from a measure of central tendency.

Measures of dispersion tell us how much the data moves away from the average

Central tendency tells average performance

Dispersion tells stability of performance

With dispersion.

Main Measures of Dispersion

- **Range**
- **Quartile Deviation**
- **Variance**
- **Standard Deviation**

Measures of Dispersion

-Range



Range

Meaning

Range is the **simplest measure of dispersion**.

It tells us how spread out the data is by looking at only:

- ☐ Maximum value
- ☒ Minimum value

Formula

Range = Maximum Value – Minimum Value

Example 1

Two cloud companies are being evaluated for hosting a university ERP system.

Company	Minimum Response Time (ms)	Maximum Response Time (ms)
Cloud A	50	450
Cloud B	120	260

Questions :

1. Calculate Range for both companies.
2. Calculate Coefficient of Range.
3. Which company is more stable?
4. If you are the IT manager, which one will you recommend and why?

Solution

Q. 1

Range=Maximum–Minimum or Range = L - S

L= Largest Value S= Smallest Value

Cloud A

Range = L - S

Range = 450–50 = 400

Range (Cloud A) = 400 ms

Cloud B

Range = L - S

Range = 260–120=140

Range (Cloud B) = 140 ms

Q.2

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

Cloud A

$$= \frac{450 - 50}{450 + 50} = \frac{400}{500} = 0.80$$

Coefficient (Cloud A) = 0.80

Cloud B

$$= \frac{260 - 120}{260 + 120} = \frac{140}{380} \approx 0.37$$

Coefficient (Cloud B) $\approx$ 0.37

Answer Q. 3

Q. 3 Which company is more stable?

Answer :

Lower value of coefficient means higher stability.

Compare:

Cloud A $\rightarrow$ 0.80

Cloud B $\rightarrow$ 0.37

Since **0.37 < 0.80**,

Cloud B is more stable.

Answer Q.4

Q. 4 If you are the IT manager, which one will you recommend and why?

Answer :

As IT Manager, I will recommend **Cloud B** because:

Cloud B has lower Range and lower Coefficient of Range, it shows less variability and hence greater stability. Therefore, Cloud B should be selected.

Example 2 :

Two data centers are tested for daily downtime (minutes) during peak examination period.

Data Center	Minimum Downtime	Maximum Downtime
DC-A	3	85
DC-B	15	40

Questions:

1. Calculate Range for both data centers.
2. Calculate Coefficient of Range.
3. Which data center shows less operational risk?
4. As CIO, which data center will you choose and why?

Q.1 Calculation of Range for Both Data Center

$$\text{Range} = L - S$$

DC-A

$$85 - 3 = 82$$

Range (DC-A) = 82 minutes

DC-B

$$40 - 15 = 25$$

Range (DC-B) = 25 minutes



Q.2 Calculation of Coefficient of Range

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

DC-A

$$\frac{85 - 3}{85 + 3} = \frac{82}{88} = 0.93$$

Coefficient (DC-A) = 0.93

DC-B

$$\frac{40 - 15}{40 + 15} = \frac{25}{55} = 0.45$$

Coefficient (DC-B) $\approx$ 0.45

Q. 3 Answer

Q. 3 Which Data Center Shows Less Operational Risk?

Answer :

Compare coefficients:

DC-A $\rightarrow$ 0.93

DC-B $\rightarrow$ 0.45

Lower coefficient means lower variability.

DC-B shows less operational risk.

Q.4 Answer

Q. 4 As CIO, Which Data Center Will You Choose and Why?

Answer :

As CIO, I will choose DC-B because:

Lower range (25 vs 82)

Lower coefficient (0.45 vs 0.93)

Lower operational uncertainty

More reliable during examination period

Conclusion

Operational risk increases when performance becomes unpredictable. DC-A's very high coefficient (0.93) indicates extreme instability.

In critical systems like university examinations, stability is more important than occasional best performance.

Example 3

Two AI-based fraud detection systems are evaluated based on response time (seconds).

System	Minimum Time	Maximum Time
AI-X	0.6	9.4
AI-Y	2.5	5.2

Questions:

1. Compute Range.
2. Compute Coefficient of Range.
3. Which system is more consistent?
4. If this is used for real-time banking transactions, which one should be selected?

Answer Q. 1

$$\text{Range} = L - S$$

AI-X

$$9.4 - 0.6 = 8.8$$

Range (AI-X) = 8.8 seconds

AI-Y

$$5.2 - 2.5 = 2.7$$

Range (AI-Y) = 2.7 seconds

Answer Q.2

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

AI-X

$$\frac{9.4 - 0.6}{9.4 + 0.6} = \frac{8.8}{10} = 0.88$$

Coefficient (AI-X) = 0.88

AI-Y

$$\frac{5.2 - 2.5}{5.2 + 2.5} = \frac{2.7}{7.7} \approx 0.35$$

Coefficient (AI-Y) $\approx$ 0.35



Q.3 Answer :

Q. 3 Which System is More Consistent?

Answer :

Compare coefficients:

AI-X $\rightarrow$ 0.88

AI-Y $\rightarrow$ 0.35

Since **0.35 < 0.88**,

AI-Y is more consistent.

Q.4 Answer :

Q. 4 If Used for Real-Time Banking Transactions, Which One Should Be Selected?

Answer :

AI-X:

Sometimes very fast (0.6 sec)

Sometimes very slow (9.4 sec)

Large fluctuation

High unpredictability

Coefficient = 0.88 (very high)

AI-Y:

Moderate but stable response

Less fluctuation

Lower operational risk

Coefficient = 0.35

Conclusion : For real-time banking transactions, AI-Y should be selected



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Example 4

A tech company monitored the **CPU utilization (%)** of its cloud servers during business hours.

CPU Usage (%)	Number of Observations
20 – 30	5
30 – 40	12
40 – 50	20
50 – 60	9
60 – 70	4

Questions:

1. Calculate the Range.
2. Calculate the Coefficient of Range.
3. Does the data indicate stable or unstable system performance?

Solution Q. 1 & 2

Smallest value = 20

Largest value = 70

Range

$$70 - 20 = 50$$

Coefficient of Range

$$\frac{70 - 20}{70 + 20} = \frac{50}{90} = 0.556$$

Answer Q. 3

Q. 3 Does the data indicate stable or unstable performance?

Coefficient of Range = 0.556

Since it is greater than 0.5, variability is moderately high.

CPU usage varies widely (from 20% to 70%).

This indicates unstable system performance.

Since CPU usage exceeds acceptable limits and variability is moderately high (Coefficient = 0.556), the system shows instability. Management should implement load optimization or infrastructure upgrades to ensure consistent performance.

Example 5

A data center recorded **daily bandwidth consumption (TB)**:

Bandwidth (TB)	Frequency
5 – 10	6
10 – 15	10
15 – 20	15
20 – 25	8
25 – 30	3

Questions:

1. Calculate the Range and the Coefficient of Range.
2. Should management purchase a higher bandwidth plan?

Solution Q. 1

Calculate the Range

$$\text{Range} = L - S$$

$$= 30 - 5$$

$$= 25 \text{ TB}$$

Calculate the Coefficient of Range

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

$$= \frac{30 - 5}{30 + 5}$$

$$= \frac{25}{35}$$

$$= 0.714$$

$$\text{Coefficient of Range} = 0.714$$

Since it is **greater than 0.5**, variability is high.



Answer Q. 2

Q. 2 Should management purchase a higher bandwidth plan?

Answer :

The range of 25 TB and coefficient of range of 0.714 indicate high variability in bandwidth usage. Since usage occasionally exceeds 25 TB, there is risk of ISP penalty. Therefore, management should upgrade bandwidth capacity or implement optimization strategies to ensure cost control and performance stability.

Measures of Dispersion

--Quartile Deviation
for Continuous data.



Meaning

Quartile Deviation measures the spread of the middle 50% of data. It tells us how much the central data values deviate from the median.

Formula

$$Q_1 = l_1 + \left(\frac{\frac{N}{4} - CF}{f} \right) \times (l_2 - l_1)$$

Where:

- l_1 = lower limit of Q_1 class
- l_2 = upper limit of Q_1 class
- N = total frequency
- CF = cumulative frequency before Q_1 class
- f = frequency of Q_1 class



Formula

$$Q_3 = l_1 + \left(\frac{\frac{3N}{4} - CF}{f} \right) \times (l_2 - l_1)$$

Where:

- l_1 = lower limit of Q_3 class
- l_2 = upper limit of Q_3 class
- N = total frequency
- CF = cumulative frequency before Q_3 class
- f = frequency of Q_3 class



Formula

Quartile Deviation:

$$QD = \frac{Q_3 - Q_1}{2}$$

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Where:

- Q_1 = First Quartile
- Q_3 = Third Quartile

Example 1

An online learning platform noticed complaints about slow video loading during peak hours.

The IT team recorded **video loading time (in seconds)** for 80 user sessions:

Loading Time (sec)	Number of Users
1 – 2	10
2 – 3	18
3 – 4	22
4 – 5	20
5 – 6	10

Questions

Q. 1 Calculate Quartile Deviation and Coefficient of QD

Q. 2 Does the Quartile Deviation indicate stable or unstable app performance?

Q. 3 Should the company:

- (a) Upgrade full server infrastructure
- (b) Optimize specific high-latency cases
- (c) Do nothing

Justify your answer using QD and Coefficient of QD.

Q. 1: Solution

$$Q_1 = \frac{N}{4} = \frac{80}{4} = 20^{th} \text{ value}$$

$$Q_3 = \frac{3N}{4} = \frac{240}{4} = 60^{th} \text{ value}$$

Using cumulative frequency:

Class	Frequency	Cumulative Frequency
1-2	10	10
2-3	18	28
3-4	22	50
4-5	20	70
5-6	10	80

Step 1: Cumulative Frequency

Class	f	CF
1–2	10	10
2–3	18	28
3–4	22	50
4–5	20	70
5–6	10	80

Step 2: Find Q_1

$$Q_1 = \frac{N}{4} = \frac{80}{4} = 20^{th} \text{ value}$$

20th value lies in class 2-3

Using formula:

$$Q_1 = l_1 + \left(\frac{\frac{N}{4} - CF}{f} \right) (l_2 - l_1)$$

Here:

- $l_1 = 2$
- $CF = 10$
- $f = 18$
- $l_2 - l_1 = 1$

$$Q_1 = 2 + \left(\frac{20 - 10}{18} \right) (1)$$

$$Q_1 = 2 + \frac{10}{18}$$

$$Q_1 = 2 + 0.56$$

$$Q_1 \approx 2.56 \text{ sec}$$

Step 3: Find Q_3

$$Q_3 = \frac{3N}{4} = \frac{240}{4} = 60^{th} \text{ value}$$

60th value lies in class 4–5

Here:

- $l_1 = 4$
- $CF = 50$
- $f = 20$
- $l_2 - l_1 = 1$

$$Q_3 = 4 + \left(\frac{60 - 50}{20} \right)$$

$$Q_3 = 4 + \frac{10}{20}$$

$$Q_3 = 4 + 0.5$$

$$Q_3 = 4.5 \text{ sec}$$



Step 4: Quartile Deviation

$$\begin{aligned} QD &= \frac{Q_3 - Q_1}{2} \\ &= \frac{4.5 - 2.56}{2} \\ &= \frac{1.94}{2} \\ QD &= 0.97 \text{ sec} \end{aligned}$$

Step 5: Coefficient of QD

$$\begin{aligned} \text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\ &= \frac{1.94}{4.5 + 2.56} \\ &= \frac{1.94}{7.06} \\ &= 0.27 \end{aligned}$$

Q. 2 : Answer

Q. 2 Does the Quartile Deviation indicate stable or unstable app performance?

Answer:

Coefficient of QD = **0.27**

Since it is close to 0 and less than 0.5:

Variability is low

Middle 50% users experience similar loading time

The app performance is relatively stable.

Q3 : Answer

Q. 3 Should the company:

- (a) Upgrade full server infrastructure
- (b) Optimize specific high-latency cases
- (c) Do nothing

Justify your answer using QD and Coefficient of QD.

Answer :

- (b) Optimize specific high-latency cases

Justification:

QD is low (0.97 sec)

Coefficient is low (0.27)

Only small group affected

Full upgrade not required

But minor optimization needed for 5–6 sec users

Correct Decision: **(b) Optimize specific high-latency cases**



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Final Managerial Decision

The Quartile Deviation and its coefficient indicate stable overall app performance. The issue affects only a small proportion of users. Therefore, targeted optimization is recommended instead of full infrastructure upgrade.

Example 2

An online payment application received complaints about delayed transaction confirmations during peak sale hours.

The IT team recorded **transaction processing time (in seconds)** for 100 transactions:

Processing Time (sec)	Number of Transactions
1 – 2	12
2 – 3	28
3 – 4	30
4 – 5	20
5 – 6	10

Questions

Q. 1 Calculate Quartile Deviation and Coefficient of QD

Q. 2 Does the Quartile Deviation indicate stable performance?

Q. 3 If management sees maximum time = 6 sec, should they panic?

Q. 4 What should the company do?

Options:

- (a) Upgrade entire infrastructure
- (b) Optimize specific slow transactions
- (c) Ignore complaints

Solution

Step 1: Cumulative Frequency

Class	f	CF
1-2	12	12
2-3	28	40
3-4	30	70
4-5	20	90
5-6	10	100

Step 2: Find Q_1

$$Q_1 = \frac{N}{4} = \frac{100}{4} = 25^{th} \text{ value}$$

25th value lies in class 2–3

Using formula:

$$Q_1 = l_1 + \left(\frac{\frac{N}{4} - CF}{f} \right) (l_2 - l_1)$$

Here:

- $l_1 = 2$
- $CF = 12$
- $f = 28$
- Class width = 1

$$Q_1 = 2 + \left(\frac{25 - 12}{28} \right)$$

$$Q_1 = 2 + \frac{13}{28}$$

$$Q_1 = 2 + 0.46$$

$$Q_1 \approx 2.46 \text{ sec}$$



Step 3: Find Q_3

$$Q_3 = \frac{3N}{4} = \frac{300}{4} = 75^{th} \text{ value}$$

75th value lies in class 4–5

Here:

- $l_1 = 4$
- $CF = 70$
- $f = 20$
- Class width = 1

$$Q_3 = 4 + \left(\frac{75 - 70}{20} \right)$$

$$Q_3 = 4 + \frac{5}{20}$$

$$Q_3 = 4 + 0.25$$

$$Q_3 = 4.25 \text{ sec}$$



Step 4: Quartile Deviation

$$QD = \frac{Q_3 - Q_1}{2}$$

$$= \frac{4.25 - 2.46}{2}$$

$$= \frac{1.79}{2}$$

$$QD = 0.895 \text{ sec}$$

Step 5: Coefficient of QD

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$= \frac{1.79}{4.25 + 2.46}$$

$$= \frac{1.79}{6.71}$$

$$= 0.27$$

Q.2 : Answer

Q. 2 Does the Quartile Deviation indicate stable performance?

Answer :

Yes.

$QD < 1 \text{ sec}$

Coefficient = 0.27 (low variability)

Transaction processing time is fairly consistent.



Q3. : Answer

Q. 3 If management sees maximum time = 6 sec, should they panic?

Answer :

No.

Only 10% transactions fall in 5–6 sec class.

Middle 50% transactions lie between:

2.46 sec and 4.25 sec.

System is generally stable.

Looking only at maximum value gives misleading conclusion.

Q. 4 : Answer

Q. 4 What should the company do?

Options:

- (a) Upgrade entire infrastructure
- (b) Optimize specific slow transactions
- (c) Ignore complaints

Answer :

Because:

Overall variability is low

Only small proportion affected

Cost-effective targeted improvement is better

Final Managerial Decision

The Quartile Deviation and its coefficient indicate stable transaction performance. While a small proportion of transactions experience delay, full infrastructure upgrade is not justified. Targeted optimization is recommended.

Example 3

A web hosting company analyzed **monthly data usage (in GB)** of 200 clients during a high-traffic period.

Data Usage (GB)	Number of Clients
100 – 200	20
200 – 300	40
300 – 400	60
400 – 500	50
500 – 600	30

Questions

Q. 1 Calculate Quartile Deviation and Coefficient of QD

Q. 2 Is the system stable?

Q3 If management only sees maximum = 600 GB, should they upgrade servers immediately?

Q4 What Should Company Do?

- (a) Upgrade full infrastructure
- (b) Monitor heavy users and introduce tiered pricing
- (c) Do nothing

Step 1: Cumulative Frequency

Class	f	CF
100–200	20	20
200–300	40	60
300–400	60	120
400–500	50	170
500–600	30	200



Step 2: Find Q_1

$$Q_1 = \frac{N}{4} = \frac{200}{4} = 50^{th} \text{ value}$$

50th value lies in class 200–300

Using formula:

$$Q_1 = l_1 + \left(\frac{\frac{N}{4} - CF}{f} \right) (l_2 - l_1)$$

$$Q_1 = 200 + \left(\frac{50 - 20}{40} \right) (100)$$

Here:

- $l_1 = 200$
- $CF = 20$
- $f = 40$
- Class width = 100

$$Q_1 = 200 + \left(\frac{30}{40} \right) (100)$$

$$Q_1 = 200 + (0.75 \times 100)$$

$$Q_1 = 200 + 75$$

$$Q_1 = 275 \text{ GB}$$

Step 3: Find Q_3

$$Q_3 = \frac{3N}{4} = \frac{600}{4} = 150^{th} \text{ value}$$

150th value lies in class 400–500

Here:

- $l_1 = 400$
- $CF = 120$
- $f = 50$
- Class width = 100

$$Q_3 = 400 + \left(\frac{150 - 120}{50} \right) (100)$$

$$Q_3 = 400 + \left(\frac{30}{50} \right) (100)$$

$$Q_3 = 400 + (0.6 \times 100)$$

$$Q_3 = 400 + 60$$

$$Q_3 = 460 \text{ GB}$$



Step 4: Quartile Deviation

$$\begin{aligned} QD &= \frac{Q_3 - Q_1}{2} \\ &= \frac{460 - 275}{2} \\ &= \frac{185}{2} \\ QD &= 92.5 \text{ GB} \end{aligned}$$

Step 5: Coefficient of QD

$$\begin{aligned} \text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\ &= \frac{185}{460 + 275} \\ &= \frac{185}{735} \\ &= 0.25 \end{aligned}$$



Q2. : Answer

Q2: Is the system stable?

Answer :

Coefficient = 0.25 (low)

Moderate spread

Usage among middle 50% clients is relatively consistent

System is reasonably stable.

Q3 : Answer

Q3: If management only sees maximum = 600 GB, should they upgrade servers immediately?

Answer :

No.

Only 30 clients fall in 500–600 class.

Middle 50% clients use between:

275 GB and 460 GB.

Maximum alone is misleading.

Q4 : Answer

Q4: What Should Company Do?

- (a) Upgrade full infrastructure
- (b) Monitor heavy users and introduce tiered pricing
- (c) Do nothing

Answer :

Correct Decision: (b)

Because:

Majority clients have moderate usage

Few heavy users drive higher consumption

Better to optimize pricing and bandwidth allocation

Final Managerial Decision :

The Quartile Deviation and its coefficient indicate moderate variability in data usage. Since only a small proportion of users consume very high bandwidth, targeted bandwidth management and pricing optimization are preferable to full infrastructure upgrade.

Example 4

A cloud service provider analyzed the **monthly data processing load (in GB)** handled by 240 enterprise clients to plan server upgrades.

Data Collected:

Monthly Processing Load (GB)	Number of Clients
100 – 300	25
300 – 500	45
500 – 700	70
700 – 900	60
900 – 1100	40

Questions

Q1. Calculate:

- (a) Quartile Deviation
- (b) Coefficient of Quartile Deviation

Q2. Does the variability indicate stable server demand or fluctuating demand?

Q3. If management upgrades infrastructure based only on the maximum load group (900–1100 GB), is that statistically justified?

Q4. Based on statistical analysis, should the company:

- (a) Upgrade servers massively
- (b) Optimize for middle-load clients
- (c) Ignore dispersion and focus only on average

Justify using statistical reasoning.



Solution

Step 1: Cumulative Frequency

Class	f	Cumulative f
100–300	25	25
300–500	45	70
500–700	70	140
700–900	60	200
900–1100	40	240

Total N = 240



Step 2: Calculate Q_1 and Q_3

$$Q_1 = \frac{N}{4} = \frac{240}{4} = 60$$

60 lies in class 300–500

$$L = 300$$

$$cf = 25$$

$$f = 45$$

$$h = 200$$

$$Q_1 = 300 + \left(\frac{60 - 25}{45} \right) \times 200$$

$$Q_1 = 300 + \left(\frac{35}{45} \right) \times 200$$

$$Q_1 = 300 + 155.56$$

$$Q_1 = 455.56 \text{ GB}$$



$$Q_3 = \frac{3N}{4} = \frac{720}{4} = 180$$

180 lies in class 700–900

$$L = 700$$

$$cf = 140$$

$$f = 60$$

$$h = 200$$

$$Q_3 = 700 + \left(\frac{180 - 140}{60} \right) \times 200$$

$$Q_3 = 700 + \left(\frac{40}{60} \right) \times 200$$

$$Q_3 = 700 + 133.33$$

$$Q_3 = 833.33 \text{ GB}$$



Q.1 Calculation of Quartile Deviation

Step 3: Quartile Deviation

$$QD = \frac{Q_3 - Q_1}{2}$$

$$QD = \frac{833.33 - 455.56}{2}$$

$$QD = \frac{377.77}{2}$$

$$QD = 188.89 \text{ GB}$$

Coefficient of QD

Step 4: Coefficient of QD

$$\begin{aligned}\text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\ &= \frac{377.77}{1288.89} \\ &= 0.293 \approx 0.29\end{aligned}$$

Answer Q.2

Q2. Does the variability indicate stable server demand or fluctuating demand?

Answer :

Coefficient of QD = 0.29 (29%)

This shows moderate dispersion.

Demand is fairly stable around middle-load clients.

Answer : Q3

Q3. If management upgrades infrastructure based only on the maximum load group (900–1100 GB), is that statistically justified?

Answer :

Only 40 out of 240 clients (16.7%) fall in highest load category.

So infrastructure decisions based only on extreme users would:

- Increase cost unnecessarily
- Ignore majority usage pattern

Statistically not justified.

Answer Q4

Q4. Based on statistical analysis, should the company:

- (a) Upgrade servers massively
- (b) Optimize for middle-load clients
- (c) Ignore dispersion and focus only on average

Justify using statistical reasoning.

Answer :

Option (b) Optimize for middle-load clients

Reason:

70 clients in 500–700 (highest frequency)

Interquartile range (455 GB to 833 GB) shows core demand range

Most revenue likely generated in mid-segment



Measures of Dispersion

- Standard Deviation
- Variance and
- Coefficient of Variation

for Continuous data.



Example 1

An IT company in Navi Mumbai monitors **server response time (in milliseconds)** during peak hours. Management wants to know whether performance is stable.

Data (Continuous Frequency Distribution)

Response Time (ms)	Number of Requests
100 – 200	8
200 – 300	15
300 – 400	22
400 – 500	10
500 – 600	5

Questions

1. Calculate Mean Response Time.
2. Calculate Variance.
3. Calculate Standard Deviation.
4. Calculate Coefficient of Variation (CV).
5. Is server performance stable? Justify using CV.
6. If competitor's CV is 18% and your CV is 28%, what decision should management take?

Solution : Q. 1 : Calculation of mean response time

Class	f	Midpoint (x)	fx
100–200	8	150	1200
200–300	15	250	3750
300–400	22	350	7700
400–500	10	450	4500
500–600	5	550	2750

$$\sum fx = 19,900$$

$$\text{Mean} = 19900/60 = 331.67 \text{ ms}$$

Solution : Q2. : Calculation of Variance

x	f	(x - mean)	(x - mean) ²	f(x - mean) ²
150	8	-181.67	33003.89	264031.12
250	15	-81.67	6670.89	100063.35
350	22	18.33	335.89	7389.58
450	10	118.33	14002.89	140028.90
550	5	218.33	47667.89	238339.45

Solution : Q. 2 & 3

$$SD = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}} = \sqrt{\frac{749852.40}{60}}$$

$$SD = \sqrt{12497.54} \quad SD \approx 111.79 \text{ ms}$$

$$\begin{aligned} \text{Variance} &= \frac{749852.40}{60} \\ &= 12497.54 \end{aligned}$$

Solution Q. 4

Coefficient of Variation (CV)

$$CV = \frac{SD}{Mean} \times 100$$

$$CV = \frac{111.8}{331.67} \times 100$$

Coefficient of Variation = 33.7%



Final Answers

- Mean Response Time = 331.67 ms
- Variance = 12,497.54
- Standard Deviation = 111.8 ms
- Coefficient of Variation = 33.7%



Answer : Q. 5

Q. 5 Is server performance stable? Justify using CV.

Answer :

$CV = 33.7\% \rightarrow$ Variation is relatively high.

This means server response time is **not very consistent**.

Even though average response time is ~ 332 ms (acceptable), fluctuations are large.

Answer : Q. 6

Q. 6 If competitor's CV is 18% and your CV is 28%, what decision should management take?

Answer :

Competitor is more stable.

Management should:

- Optimize load balancing
- Upgrade server hardware
- Improve caching/indexing
- Monitor peak traffic bottlenecks

Example 2

An MCA-based software firm measures **bug fixing time (in minutes)** for reported issues

Resolution Time (mins)	Number of Bugs
0 – 30	10
30 – 60	18
60 – 90	25
90 – 120	12
120 – 150	5

Questions:

Questions:

1. Compute Mean resolution time.
2. Compute Variance and SD.
3. Compute CV.
4. If management wants faster & consistent service, what does SD suggest?

Q. 1 Solution : Calculation of mean

Class	f	Midpoint (x)	fx
0–30	10	15	150
30–60	18	45	810
60–90	25	75	1875
90–120	12	105	1260
120–150	5	135	675

$$\sum fx = 4770$$

$$\text{Mean} = 4770/70 = 68.14 \text{ minutes}$$

Solution: Q. 2 : Calculation of Variance

x	f	(x – 68.14)	(x – 68.14) ²	f(x – mean) ²
15	10	-53.14	2823.86	28238.6
45	18	-23.14	535.45	9638.1
75	25	6.86	47.06	1176.5
105	12	36.86	1358.66	16303.9
135	5	66.86	4470.26	22351.3

Solution: Q. 2 : Calculation of Variance

$$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{N}$$

$$\sum f(x - \bar{x})^2 \approx 77,708.4$$

$$\text{Variance} = 77708.4/70$$

$$\text{Variance} \approx 1,110.12$$

Step 4: Standard Deviation

$$\sigma = \sqrt{1110.12}$$

$$\text{SD} \approx 33.32 \text{ minutes}$$

Step 5: Coefficient of Variation (CV)

$$CV = \frac{SD}{Mean} \times 100$$

$$CV = \frac{33.32}{68.14} \times 100$$



Final Answers

- Mean Resolution Time = 68.14 minutes
- Variance $\approx 1,110.12$
- Standard Deviation ≈ 33.32 minutes
- Coefficient of Variation $\approx 48.9\%$



Decision Making

- Mean shows average efficiency
- SD shows consistency
- CV helps compare with competitor firms

If competitor CV = 25%

Your CV = 49%

Competitor is more process-stable.

Answer Q.4

Q. 4 If management wants faster & consistent service, what does SD suggest?

Answer :

Mean response time = 68.14

Standard Deviation = 33.32

SD is almost 49% of the mean, which is relatively high.

What does this suggest?

Response times are widely spread

Service is not very consistent

Some requests are much slower than average

If management wants faster & consistent service:

- They should try to reduce variability
- Optimize server load handling
- Improve infrastructure or balancing
- Aim to reduce SD closer to 10–15 range

Conclusion:

Current SD suggests moderate to high variability, so performance is not fully stable.

Thank You



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